

ZAP-FIX Latency Model: Formal Bounds

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Abstract

This document formalises the latency model claimed in the ZAP-FIX paper. We give a per-message round-trip decomposition, derive a steady-state per-message bound (Theorem 1), a session-resumption bound (Theorem 2), and a sequence-determinism lemma. Concrete numerical budgets are tabulated so that operators can calibrate the model against their colocated environment.

1 Notation and Model

Definition 1 (Per-message round trip). *For two FIX peers A and B on the same session and a FIX application message M originating at A , the round-trip latency $T_{\text{rtt}}(M)$ is the wall-clock interval from the moment A 's application layer hands M to its FIX session library to the moment A 's application layer observes the corresponding acknowledgement message M' (typically an *ExecutionReport-New* (150=0)) returned by B .*

We decompose

$$T_{\text{rtt}}(M) = T_{\text{ser}}^A(M) + T_{\text{tx}}^A(M) + T_{\text{net}}^{AB}(M) + T_{\text{rx}}^B(M) + T_{\text{parse}}^B(M) + T_{\text{app}}^B(M) + T_{\text{rev}}^{BA}(M') \quad (1)$$

where the components are: serialisation (T_{ser}); kernel transmit including NIC posting (T_{tx}); on-wire and switch fabric (T_{net}); kernel receive including NIC delivery (T_{rx}); session-layer parse (T_{parse}); application processing including matching, risk, and acknowledgement formation (T_{app}); and the symmetric reverse-path total (T_{rev}).

Definition 2 (Transport binding). *A transport binding Π for FIX is a triple $(\sigma_{\Pi}, \pi_{\Pi}, \kappa_{\Pi})$ where σ_{Π} is a serialiser $M \rightarrow \text{bytes}$, π_{Π} is the parser $\text{bytes} \rightarrow M$, and κ_{Π} is the channel cost: a function from message size to (kernel, NIC, fabric) cost.*

We compare Π_{tls} (TCP+TLS-FIX, SOH text body) with Π_{zap} (ZAP-FIX, envelope; text or binary body mode).

2 Steady-State Per-Message Bound

Lemma 1 (Channel cost monotonicity). *For any reliable byte-oriented or frame-oriented channel and a message of size s bytes,*

$$\kappa(s) = \kappa_0 + \kappa_1 \cdot s$$

with κ_0 a per-message constant (kernel call, NIC doorbell, fabric arbitration) and κ_1 a per-byte constant (serialisation onto wire). Both are non-negative.

The lemma is standard; it captures the empirical observation that for messages well under one MTU, κ_0 dominates $\kappa_1 \cdot s$. We use it only to argue monotonicity in s .

Theorem 1 (Steady-state per-message latency). *Let M be a FIX message exchanged in steady state (no handshake, no resend). Let $s_\Pi(M) = |\sigma_\Pi(M)|$. Define $\Delta_{\text{ser}}(M) = T_{\text{ser}}^{\Pi_{\text{tls}}}(M) - T_{\text{ser}}^{\Pi_{\text{zap}}}(M)$ and analogously $\Delta_{\text{parse}}(M)$. Then*

$$T_{\text{rtt}}^{\Pi_{\text{zap}}}(M) \leq T_{\text{rtt}}^{\Pi_{\text{tls}}}(M) - (\Delta_{\text{ser}}(M) + \Delta_{\text{parse}}(M)) - \kappa_1 \cdot (s_{\Pi_{\text{tls}}}(M) - s_{\Pi_{\text{zap}}}(M)).$$

Proof. By Equation (??), the only components that differ between Π_{tls} and Π_{zap} in steady state are T_{ser} , T_{parse} , and the channel contribution $T_{\text{tx}} + T_{\text{net}} + T_{\text{rx}}$, which by Lemma ?? differs by exactly $\kappa_1 \cdot \Delta s$, with $\Delta s = s_{\Pi_{\text{tls}}}(M) - s_{\Pi_{\text{zap}}}(M)$. The κ_0 contribution is identical because both bindings emit exactly one message-bearing unit (one TLS record carrying one FIX message versus one ZAP frame carrying one FIX message). Substituting into (??) yields the claim. $\square$

Concrete numerical bound. Table ?? substitutes representative values for an optimised colocated client. Values are in nanoseconds.

Component	TCP+TLS-FIX	ZAP-FIX (text)	ZAP-FIX (binary)
Serialise (<code>NewOrderSingle</code>)	1500–3000	1600–3100	200–500
Kernel transmit	300–800	300–800	300–800
Fabric one-way (1×10G, 1 hop)	600–900	600–900	600–900
Bytes on wire	220	240	130
Per-byte cost contribution	35–90	38–96	21–52
Kernel receive	300–800	300–800	300–800
Parse (SOH walk <i>vs.</i> envelope)	800–1500	100–250	50–150
Application matching (typical)	5000–15000	5000–15000	5000–15000
Reverse-path symmetric	$\approx$ above	$\approx$ above	$\approx$ above
Total round trip T_{rtt}	50000–200000	50000–200000	47000–195000

Table 1: Indicative per-component latency budget (ns) for a `NewOrderSingle/ExecutionReport-New` round trip. The transport-attributable difference $\Delta_{\text{ser}} + \Delta_{\text{parse}} + \kappa_1 \Delta s$ is roughly 1.5–3 μs (text mode improvement on parse alone) and roughly 3–8 μs (binary mode), against a colocated budget of 50–200 μs . In multi-hop or non-colocated deployments the byte-count saving compounds.

3 Session-Resumption Bound

We model session resumption as the wall-clock interval from the moment a peer detects an unhealthy transport (TCP RST, missed heartbeat, pod migration signal) to the moment the FIX session is back in normal flow with no application-visible gap.

Definition 3 (Resumption time). T_{resume}^{Π} is the time from disconnection detection t_0 to the moment t_1 at which both peers have re-established a session and the in-flight gap (if any) has been closed by `FIX ResendRequest/SequenceReset`.

For TCP+TLS-FIX,

$$T_{\text{resume}}^{\Pi_{\text{tls}}} = T_{\text{tcp}} + T_{\text{tls}} + T_{\text{logon}} + T_{\text{gapdetect}}^{\text{stream}} + T_{\text{resend}} + T_{\text{backoff}} \quad (2)$$

where T_{tcp} is one round trip plus possible retransmit backoff after the loss event, T_{tls} is one round trip for TLS 1.3 1-RTT (or 0-RTT resumption with replay-window risk), $T_{\text{gapdetect}}^{\text{stream}}$ is the cost of byte-stream scanning to locate FIX message boundaries, and T_{backoff} accumulates middle-box and retry backoffs.

For ZAP-FIX,

$$T_{\text{resume}}^{\Pi_{\text{zap}}} = T_{\text{zap}} + T_{\text{logon}} + T_{\text{gapdetect}}^{\text{frame}} + T_{\text{resend}} \quad (3)$$

where T_{zap} is the single-RTT ZAP handshake (with post-quantum hybrid KEM) and $T_{\text{gapdetect}}^{\text{frame}} = O(1)$ because the next envelope header carries the next `MsgSeqNum` explicitly.

Theorem 2 (Session-resumption bound). *There exist $C_1, C_2 > 0$, dependent only on the deployed colocation environment, such that*

$$T_{\text{resume}}^{\Pi_{\text{zap}}} \leq T_{\text{rtt-net}} + T_{\text{logon}} + C_1 + C_2 \cdot N_{\text{gap}}$$

where $T_{\text{rtt-net}}$ is one network round trip and N_{gap} is the number of FIX messages in the gap. By contrast,

$$T_{\text{resume}}^{\Pi_{\text{tls}}} \geq 2 \cdot T_{\text{rtt-net}} + T_{\text{logon}} + C'_1 + C'_2 \cdot N_{\text{gap}} + T_{\text{backoff}}$$

with $C'_1 \geq C_1$ and $C'_2 \geq C_2$, and T_{backoff} non-negative.

Proof. The ZAP bound follows from (??): T_{zap} contributes one round trip; gap detection is $O(1)$ per missed message because each frame’s envelope header is independently addressable (set C_1 to cover the constant per-frame cost and C_2 to cover the per-resent-message cost). The TLS bound follows from (??): TCP and TLS each contribute one round trip; byte-stream scanning costs at least the per-frame cost of ZAP and typically more because boundary detection requires parsing each message’s length-or-checksum field; T_{backoff} is non-negative and is incurred whenever the disconnect cause is loss-driven rather than peer-initiated. $\square$

Corollary. For pod-migration scenarios common in containerised venue infrastructure, where a TCP connection is unavoidably terminated even when both endpoints survive, ZAP-FIX recovers in approximately $T_{\text{rtt-net}} + T_{\text{logon}}$, which on a 1 ms-RTT wide-area path is on the order of 1–2 ms. TCP+TLS-FIX recovery on the same path includes TCP three-way handshake plus TLS key exchange plus session-state reconstruction; the dominant residual cost is T_{backoff} , which can be tens of milliseconds in the presence of competing flows.

4 Order-of-Fills Determinism

Lemma 2 (Sequence preservation). *For any two consecutive FIX application messages M_i, M_{i+1} emitted on a ZAP-FIX session by the same peer, the corresponding ZAP frames $F(M_i), F(M_{i+1})$ satisfy:*

1. $F(M_i)$ is delivered before $F(M_{i+1})$ at the receiver.
2. $\text{MsgSeqNum}(F(M_{i+1})) = \text{MsgSeqNum}(F(M_i)) + 1$.

Proof. Property (i) is the in-order delivery guarantee of the ZAP session binding: ZAP frames on a single session form a totally ordered stream, and the send-side serialiser emits one frame per message without coalescing or reordering. Property (ii) follows from the construction of Section 3.3 of the paper, which fixes the `MsgSeqNum` field of the envelope header to the FIX session `MsgSeqNum` at emission time and refuses out-of-order emission. $\square$

Corollary 1 (Order-of-fills determinism). *A matching engine consuming application messages from a ZAP-FIX session observes the same per-session arrival order as it would under a TCP+TLS-FIX binding with identical wire conditions. The FIX session-layer invariant in Equation (1) of the paper is preserved.*

5 Threats Outside the Model

The bounds above hold against:

- Loss-driven TCP recovery (TLS-FIX pays T_{backoff} that ZAP-FIX does not, given a stateless re-handshake).
- Cryptographic harvest-now-decrypt-later: ZAP-FIX inherits post-quantum hybrid KEM from the underlying ZAP transport; TLS 1.3 in deployed profiles does not.
- CompID-spoofing at session establishment: ZAP-FIX binds the FIX `SenderCompID` to the ZAP identity at handshake time, and the gateway rejects mismatches; TLS-FIX with client-cert authentication achieves a comparable bind, but the deployed default is server-only TLS plus FIX-layer password.

The bounds do *not* address:

- Application-layer matching cost T_{app} , which is unaffected by the transport choice.
- Cross-venue race conditions, which are governed by venue-side scheduling rather than the institutional channel.
- Adversarial NIC behaviour or hardware-clock skew exceeding the RTS 25 envelope; both bindings depend on the underlying PTP discipline.

6 Summary

Theorem ?? establishes that, in steady state and for every FIX message M , ZAP-FIX per-message latency is no greater than TCP+TLS-FIX latency, and is strictly less by the serialisation, parse, and per-byte savings tabulated above. Theorem ?? establishes that ZAP-FIX session resumption is bounded above by a single network round trip plus a FIX Logon, regardless of pod migration or partition, while TCP+TLS-FIX pays at least two round trips plus loss-driven backoff. Lemma ?? establishes that ZAP-FIX preserves the FIX session-layer determinism that all downstream order-handling and audit logic depend on.